

- 188.** Let $N = 3 \times 6 \times 9 \times 12 \times \dots \times 102 = 3^{34} \times (1 \times 2 \times 3 \times 4 \times \dots \times 34) = 3^{34} \times 34!$
Clearly, highest power of 2 in $34!$ is much greater than that of 5.
So, number of zeros in $N =$ Highest power of 5 in $34! = \left[\frac{34}{5} \right] + \left[\frac{34}{5^2} \right] = 6 + 1 = 7$.
- 189.** 3^4 gives unit digit 1. So, $(3^4)^{500}$ gives unit digit 1.
And, 3^3 gives unit digit 7.
 $\therefore (13)^{2003}$ gives unit digit $= (1 \times 7) = 7$.
- 190.** 3^4 gives unit digit 1. So, $(3^4)^{24}$ gives unit digit 1.
And, 3^3 gives unit digit 7.
 $\therefore 3^{99} = (3^4)^{24} \times 3^3$ gives unit digit (1×7) i.e. 7.
- 191.** (A) 7^4 gives unit digit 1. So, $7^{16} = (7^4)^4$ gives unit digit 1.
 $\therefore (1827)^{16}$ gives unit digit 1. So, $A \rightarrow (1)$.
(B) 3^4 gives unit digit 1. So, $(3^4)^4$ gives unit digit 1.
 $\therefore 3^{19} = 3^{16} \times 3^3$ gives unit digit $= (1 \times 7) = 7$.
 $\therefore (2153)^{19}$ gives unit digit 7. So, $B \rightarrow (4)$.
(C) 9^2 gives unit digit 1. So, $(9^2)^{10}$ gives unit digit 1.
 $\therefore 9^{21} = (9^2 \times 9)$ gives unit digit $= (1 \times 9) = 9$.
 $\therefore (5129)^{21}$ gives unit digit $= 9$. So, $C \rightarrow (5)$.
Hence, $A \quad B \quad C$ is the correct result.
 $\quad \quad \quad 1 \quad 4 \quad 5$
- 192.** Unit digit of $(67)^{25} =$ Unit digit of 7^{25} .
Unit digit of 7^4 is 1 and so the unit digit of $(7^4)^6$ is 1.
 $\therefore$ Unit digit of $7^{25} = (1 \times 7) = 7$.
Hence, the unit digit of $(7^{25} - 1)$ is $(7 - 1) = 6$.
- 193.** Unit digit in the given product $=$ Unit digit of $(4 \times 8 \times 7 \times 3)$, which is 2.
- 194.** Unit digit in the given product $= 8$.
Unit digit of $(9 \times 6 \times x \times 4)$ is 8. So, $x = 3$.
- 195.** Unit digit of 7^4 is 1. So, the unit digit of $(7^4)^{23}$ is 1.
 $\therefore$ Unit digit of $7^{95} =$ Unit digit of $(7^{92} \times 7^3) = (1 \times 3) = 3$.
Unit digit of 3^4 is 1. So, the unit digit of $(3^4)^{14}$ is 1.
 $\therefore$ Unit digit of $3^{58} =$ Unit digit of $(3^{56} \times 3^2) = (1 \times 9) = 9$.
Hence, the unit digit of $(7^{95} - 3^{58}) = (3 - 9) = 4$.
- 196.** Unit digit of 4^2 is 6. So, unit digit in $(4^2)^{63}$ is 6.
 $\therefore$ Unit digit of $(784)^{126} =$ Unit digit in 4^{126} , which is 6.
Unit digit in $4^{127} =$ Unit digit in $(4^{126} \times 4) =$ Unit digit in (6×4) , which is 4.
 $\therefore$ Unit digit in $(784)^{127}$ is 4.
Hence, unit digit of $\{(784)^{126} + (784)^{127}\} =$ Unit digit of $(6 + 4) =$ Unit digit of 10, which is 0.
- 197.** Unit digit in $[(251)^{98} + (21)^{29} - (106)^{100} + (705)^{35} - (16)^4 + 259] =$ Unit digit in $(1 + 1 - 6 + 5 - 6 + 9) = 4$.
- 198.** Unit digit in the given product
 $=$ Unit digit in $(4^{1793} \times 5^{317} \times 1^{491})$
Unit digit in 4^2 is 6 and so the unit digit in $(4^2)^{896}$ is 6.
 $\therefore$ Unit digit of $4^{1793} =$ Unit digit in $(6 \times 4) =$ Unit digit in 24, which is 4.
Unit digit in 5^{317} is 5 and the unit digit in 1^{491} is 1.
 $\therefore$ Unit digit in the given product $=$ Unit digit in $(4 \times 5 \times 1)$, which is 0.
- 199.** Let $x = 2y$. Then, $x^{4n} = (2y)^{4n} = \{(2y)^4\}^n = (16y^4)^n$.
 $y = 1, 2, 3, 4, 5, 6, 7, 8, 9$ gives unit digit as 6 in $(16y^4)^n$.
But, $y = 5$, gives unit digit 0 in $(16y^4)^n$.
Hence, the unit digit is 0 or 6.
- 200.** In 5^n we have 5 as unit digit and in 6^m we have 6 as unit digit.
 $\therefore$ Unit digit in $(5^n + 6^m) =$ Unit digit in $(5 + 6) =$ Unit digit in 11 $= 1$.
- 201.** We have: $(2^{12n} - 6^{4n}) = (2^{12n} - 2^{4n} \times 3^{4n}) = 2^{4n} (2^{8n} - 3^{4n})$.
Putting $n = 1$, we get the number $2^4 (2^8 - 3^4)$
 $= 16 (256 - 81) = (16 \times 175) = 2800$
Hence, the number formed by last two digits is 00.
- 202.** $\left(\frac{1}{5}\right)^{2000} = (0.2)^{2000}$.
Last digit of $(0.2)^{2000} =$ Last digit of $(0.2)^4 = 6$.
- 203.** Sum of digits in the two numbers $= 19 + 15 = 34$.
So, the product will have 33 or 34 digits.
Since $36 \times 34 = 1224$ (i.e. product has $2 + 2 = 4$ digits), so the number of digits in x is 34.
- 204.** We know that a number having x, y, z as its digits, is a multiple of 11, if $z + x - y = 0$
Hence, $y = z + x$.
- 205.** $6135n2$ is divisible by 9 if $(6 + 1 + 3 + 5 + n + 2) = (17 + n)$ is divisible by 9.
This happens when the least value of n is 1.
- 206.** In 978626, we have $(6 + 6 + 7) - (2 + 8 + 9) = 0$.
Hence, 978626 is completely divisible by 11.
- 207.** Sum of all digits $= 12$, which is divisible by 3. So, the given number is divisible by 3.
(Sum of digits at odd places) $-$ (Sum of digits at even places) $= 6 - 6 = 0$.
So, the given number is divisible by 11.
The given number when divided by 37 gives 3003003003.
So, the given number is divisible by 37.
The given number when divided by 111 gives 1001001001.
Clearly, it is divisible by 111 as well as by 1001.
Hence, the given number is divisible by each one of 3, 11, 37, 111 and 1001.
- 208.** We have $18 = 2 \times 9$, where 2 and 9 are co-primes.
But, 65043 is not divisible by 2. So, it is not divisible by 18.
- 209.** Let the missing digit be x . Then, $(80 + x)$ must be divisible by 4. Hence, $x = 4$.
- 210.** Sum of the digits in 4006020 is 12, which is divisible by 3. Hence, 4006020 is divisible by 3.
- 211.** Clearly, (d) is true.
- 212.** For 4823718 we have
 $(8 + 7 + 2 + 4) - (1 + 3 + 8) = (21 - 12) = 9$, which is not a multiple of 11.
 $\therefore$ 4823718 is not divisible by 11.
Consider the number 4832718.
We have $(8 + 7 + 3 + 4) - (1 + 2 + 8) = (22 - 11) = 11$, which is a multiple of 11.
Hence, 4832718 is divisible by 11.
- 213.** Consider the number 17325.
Its unit digit is 5. So, it is divisible by 5.
Sum of its digits $= (5 + 2 + 3 + 7 + 1) = 18$, which is divisible by 3.
So, the given number is divisible by 3.
And, since 5 and 3 are co-primes, so the given number is divisible by (5×3) , i.e. 15.
- 214.** Given number is 7386038.
Sum of its digits $= 35$, which is not divisible by any of 3 and 9.
So, the given number is not divisible by any of 3 and 9.
Also, 38 is not divisible by 4. So, the given number is not divisible by 4.